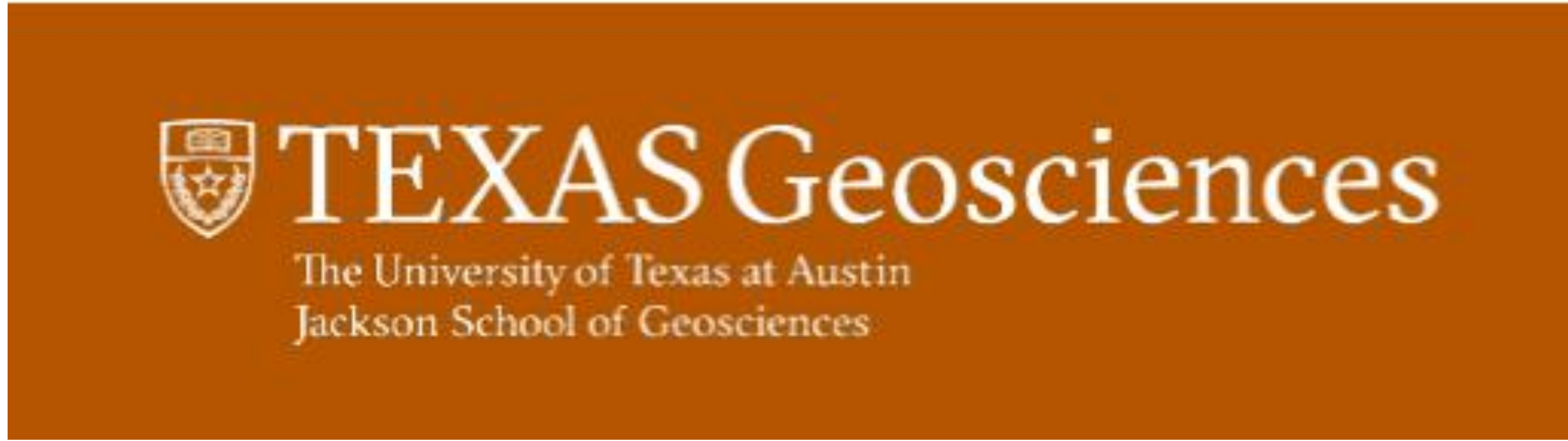


Integrating Satellite Altimetry and GRACE Gravimetry to Evaluate Reservoir Storage Variations

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Introduction

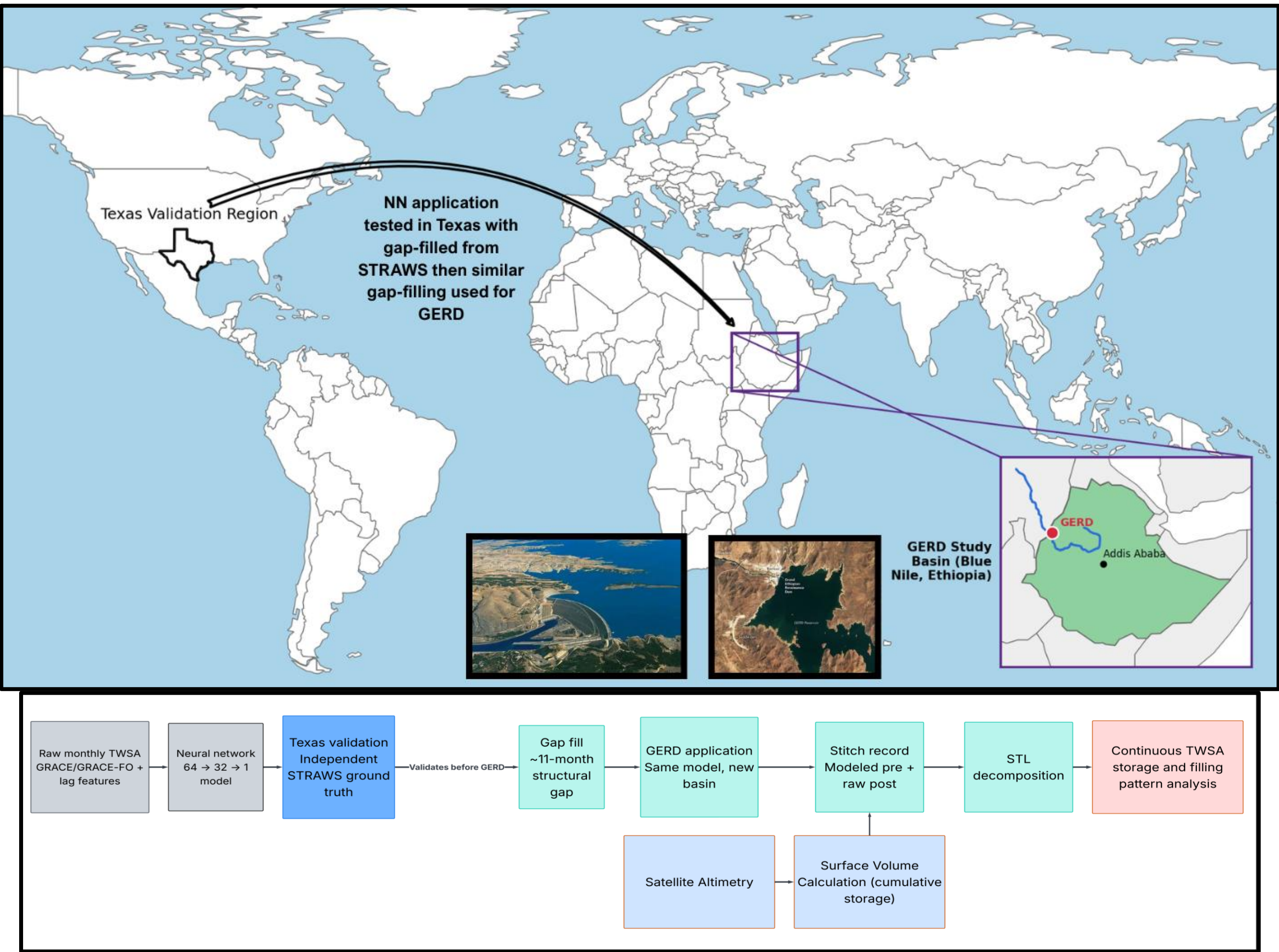
- Reservoir seepage and storage changes are hard to monitor directly, especially in transboundary basins where ground instrumentation is limited or politically restricted
- GRACE/GRACE-FO satellites track monthly total water storage changes across a basin, but the record has gaps — notably an 11-month gap between missions that breaks continuity during critical periods like active reservoir filling
- This project fills those gaps using a neural network trained on the record's trend and seasonal patterns, building on prior methods (Longuevergne et al., 2013) for isolating reservoir signals from GRACE data
- Satellite altimetry provides independent water level and surface area measurements, used to validate and constrain the coarser GRACE mass anomalies
- Demonstrated on the Grand Ethiopian Renaissance Dam (GERD), a rapidly filling reservoir central to the ongoing Nile water dispute, where an independent, continuous storage record matters for real-world monitoring

Research Goal

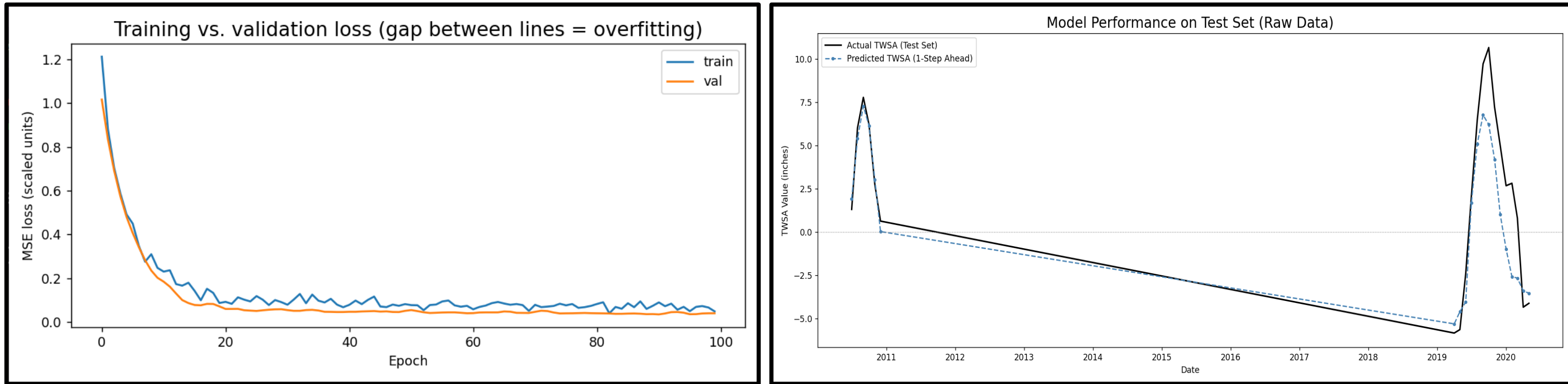
- Develop a satellite-based framework that produces a continuous, gap-free water storage record for GERD by combining GRACE gravimetry and satellite altimetry
- Use this record to characterize the timing, rate, and pattern of reservoir filling in a transboundary basin where independent monitoring is critical

Methods

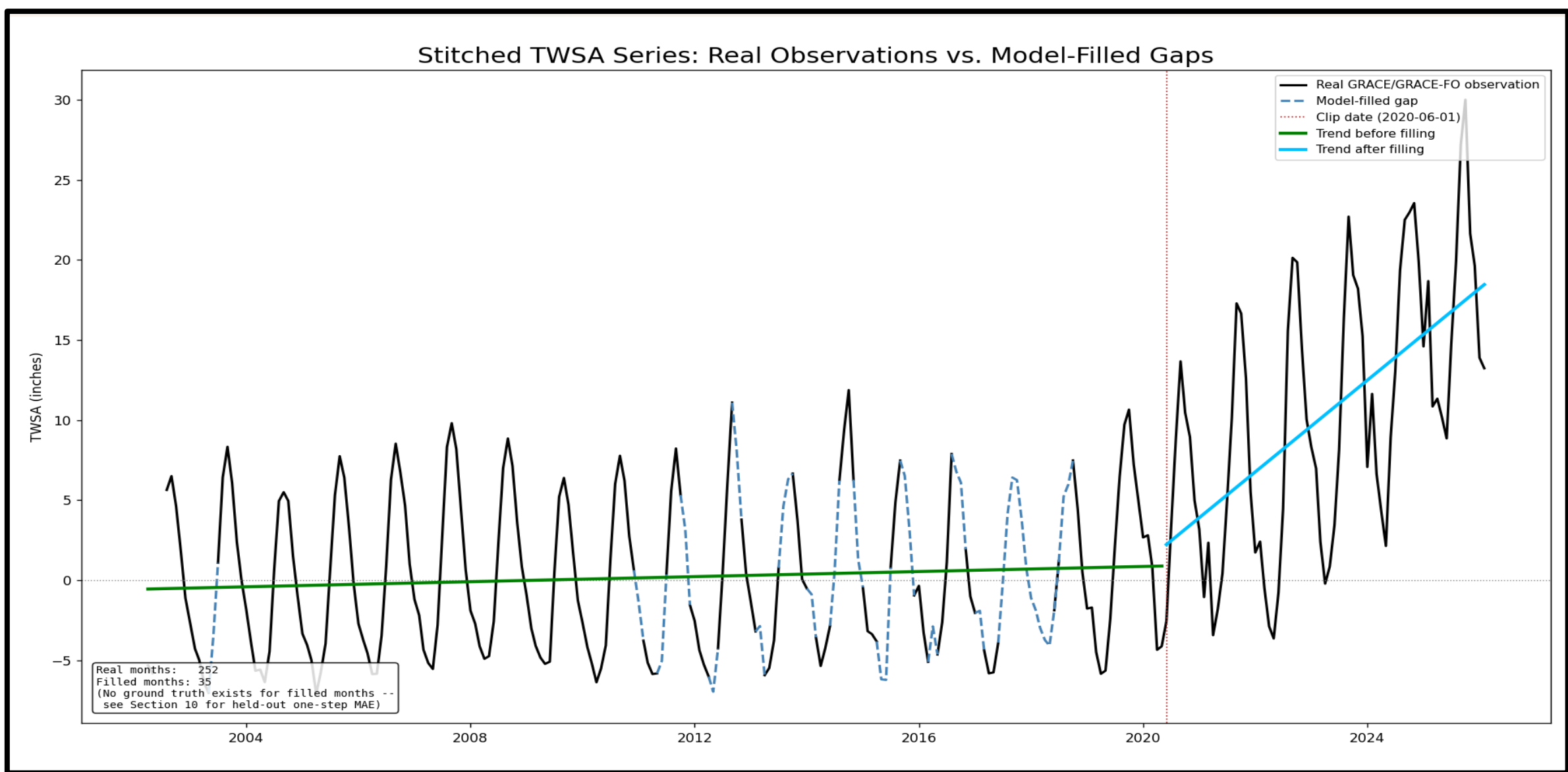
- Trained a neural network on lag features of GRACE/GRACE-FO TWSA data to fill gaps, validating it against STRAWS data in Texas before applying it to GERD
- Trained the model on pre-GERD info then stitched the gap-free data record back on to produce one continuous TWSA series used for seepage and storage-change analysis



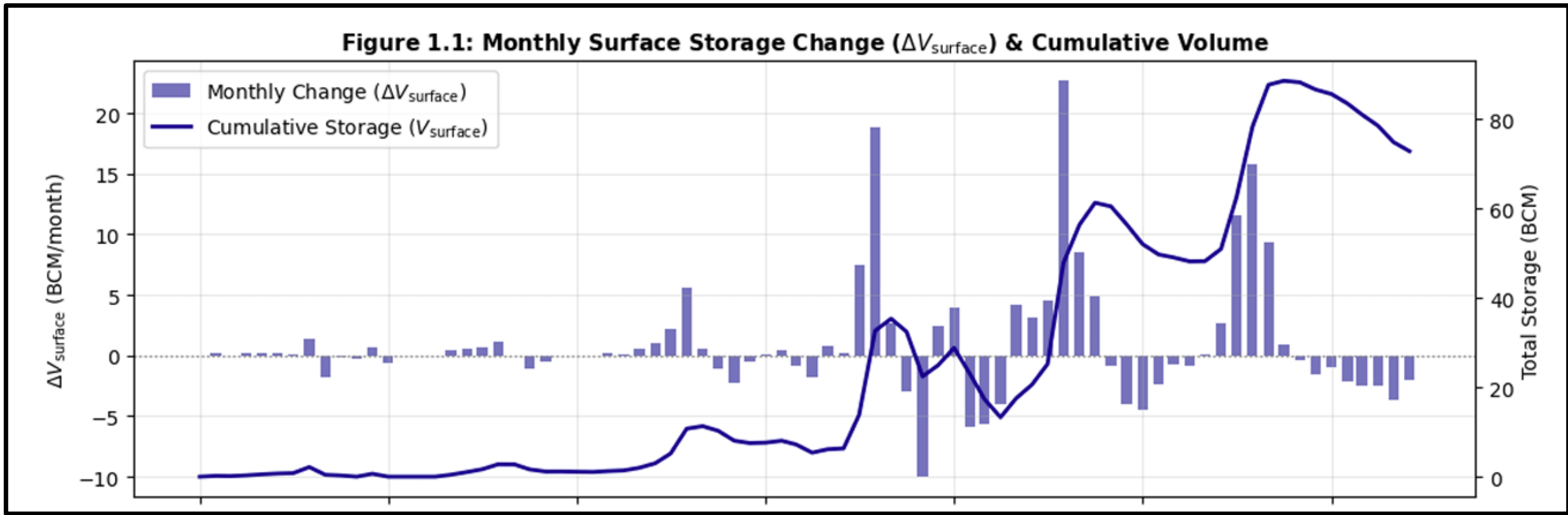
Figures and Results



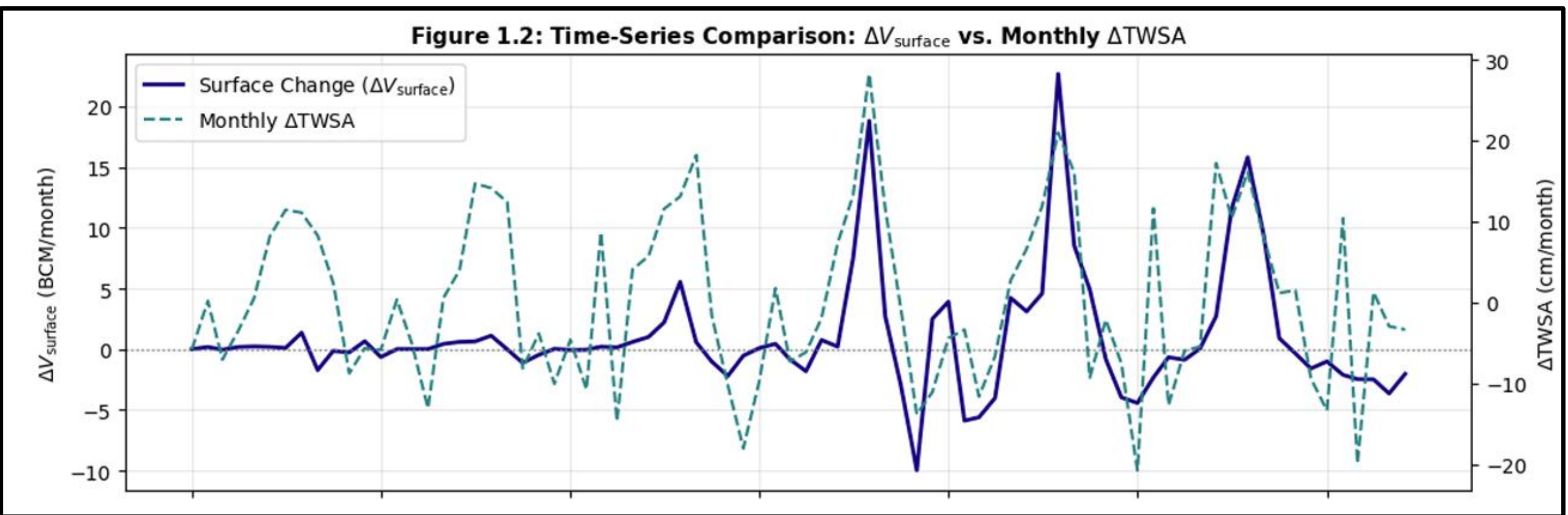
- Train/val loss converges over 100 epochs with no overfitting, model closely tracks actual GERD TWSA in one-step-ahead tests
- This measures one-step error only, recursive gap-filling used for GERD gap can accumulate error



- Reconstructed 2003–2025 TWSA record (real + model-filled gaps) shows a clear, unmodeled upward shift after July 2020, confirming reservoir accumulation



- Peak surface storage reached 88.61 BCM (October 2024)
- Annual filling rate rose sharply: +6.35 BCM/yr (2021) → +17.42 (2022) → +31.58, the peak year (2023) → +30.15 (2024)
- Highest single-month impoundment: +22.68 BCM in August 2023



- Monthly surface change correlates with monthly Δ TWSA at $r = 0.61$ during July–September monsoon months
- Cumulative surface volume aligns with total TWSA anomalies at $r = 0.70$ across multi-year filling stages

Conclusion

- Neural network successfully reconstructed missing TWSA values in both the Texas validation case and the GERD record, producing a **continuous, gap-free time series** that independently captures the onset of reservoir filling in 2020
- Reservoir reached a peak surface storage of **88.61 BCM in October 2024**, with annual filling accelerating from +6.35 BCM/yr (2021) to a peak of **+31.58 BCM/yr (2023)**, including a single-month high of **+22.68 BCM in August 2023**
- Impoundment is monsoon-synchronized, filling tracks heavy July–September rainy seasons rather than occurring continuously year-round
- Altimetry-derived surface volume and GRACE TWSA show strong agreement ($r = 0.61$ seasonally, $r = 0.70$ across multi-year trends), confirming both satellite methods are capturing the same hydrological filling events
- Rapid surface-volume spikes during peak filling months suggest water is also moving laterally into **bank storage** (surrounding unsaturated rock and soil), not just accumulating in the reservoir
- Isolating a seepage-specific signal from basin-wide GRACE data proved limited by GRACE's spatial resolution relative to the reservoir's footprint
- Future work: explore downscaling methods or complementary local measurements to disaggregate the dam-scale signal from broader basin hydrology

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